



Classification, registration and segmentation of ear canal impressions using convolutional neural networks

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ABSTRACT

Today, fitting bespoke hearing aids involves injecting silicone into patients' ears to produce ear canal molds. These are subsequently 3D scanned to create digital ear canal impressions. However, before digital impressions can be used they require a substantial amount of effort in manual 3D editing. In this article, we present computational methods to pre-process ear canal impressions. The aim is to create automation tools to assist the hearing aid design, manufacturing and fitting processes as well as normalizing anatomical data to assist the study of the outer ear canal's morphology. The methods include classifying the handedness of the impression into left and right ear types, orienting the geometries onto the same coordinate system sense, and removing extraneous artifacts introduced by the silicone mold. We investigate the use of convolutional neural networks for performing these semantic tasks and evaluate their accuracy using a dataset of 3000 ear canal impressions. The neural networks proved highly effective at performing these tasks with 95.8% adjusted accuracy in classification, 92.3% within 20° angular error in registration and 93.4% intersection over union in segmentation.

1. Introduction

Hearing loss is an important public healthcare problem that affects the quality of life for 1.5 Billion people around the Globe. Those suffering from hearing loss have an increased risk of dementia. These problems are expected to worsen with the globally aging population. It is not surprising, therefore, that improved access to hearing aids is one of the key actions recommended by the World Health Organization (WHO, 2018). The morphology of the outer ear canal is not only highly important for a good hearing aid fit (Paulsen et al., 2002), but also essential for a wide range of scientific and clinical purposes (Voss et al., 2020). There has been, however, limited data available that can be used to understand the ear's anatomy and variations among individuals (Voss et al., 2020). Fortunately, the advent of technologies such as 3D scanning, computer-aided design and additive manufacturing transformed the design, production and fitting of hearing aids (Valente et al., 1998) simplifying the measurement of the ear canal's cavity (Pirzanski, 2006), integrating electrical and mechanical components, and 3D printing, rather than casting, bespoke in-the-ear shells (Sickel et al., 2011; Baloch et al., 2010). Nevertheless, creating a digital model still requires injecting silicone into the ear canal to produce an Ear Canal Impression (ECI) which is afterward digitized using a 3D scanner (Sullivan, 2007).

Today it is common for audiologists to use 3D scanners to create digital models of ECIs and large volumes of human ear canal scans now exist. This offers an opportunity to study the ear canal's morphology using computational methods. Indicatively, 10,000 ECIs were available here, whereas in past works only a few dozen (Stinson and Lawton, 1989; Paulsen et al., 2004) to low hundreds were examined (Baloch et al., 2010; Voss et al., 2020). However, before the data can be useful they must be substantially 3D edited. Indicatively, only 3000 ECIs were used due to the manual labor required to pre-process and normalize the data. Pre-processing ECIs presents two major challenges that stem from (a) the ambiguity of associating anatomical semantics with geometric features, and (b) the discrepancies introduced by creating and digitizing the silicone impressions.

While the anatomy of the outer ear has been extensively studied in the past (Alvord and Farmer, 1997; Møller, 2012) and regions of the ear's surface have been qualitatively attributed, translating those descriptions into quantifiable geometric features is non-trivial (Baloch et al., 2010). Additionally, inconsistencies in the silicone injection process, and modifications to the ear molds or 3D scans thereof, introduce substantial variations among ECIs. For instance, the scanned ear canal's length is affected by the insertion depth of a cotton block, which is

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used to protect the patient's eardrum, before injecting the silicone. Moreover, the boundary of the external ear surface is also determined by the volume of silicone used. Excessive silicone is often removed from ECIs by the clinician with a knife, creating surfaces irrelevant to the ear's morphology. It is thus difficult to unambiguously discern the parts of the 3D geometry that were in contact with the ear canal's surface. Finally, 3D scanners use sensor-oriented coordinate systems and the placement of ECIs within these devices is often inconsistent. The objective of this work is to overcome the challenge of ECI data pre-processing aiming to assist hearing aid designers, by offering tools that can reduce manual geometry 3D editing; audiologists, by providing coherent and conveniently visualized representations of the ear canal and data scientists, by offering tools for normalizing large datasets.

2. Related work

The outer ear canal's shape has been studied in the context of hearing aid design, with the emphasis on acoustical performance (Voss et al., 2020; Stinson and Lawton, 1989; Farmer-Fedor and Rabbitt, 2002; Stinson and Daigle, 2005), design and manufacturing considerations (Paulsen et al., 2002; Baloch et al., 2010; Paulsen et al., 2004; Unal et al., 2008; Paulsen and Hilger, 2003; Slabaugh et al., 2008), as well as comfort characteristics (Sullivan, 2007; Unal et al., 2008). Outer ear canal 3D images have been acquired using contact scanning sensors (Stinson and Lawton, 1989), computer tomography (CT) (Egolf et al., 1993), and more commonly today laser scanning (Paulsen et al., 2002; Pirzanski, 2006; Sullivan, 2007; Farmer-Fedor and Rabbitt, 2002; Voss et al., 2020). Detecting anatomical features and semantic annotation of the ear canal's geometry has been approached using expert systems (Sickel et al., 2011), principal components analysis (Baloch et al., 2010), active shape modeling (Paulsen et al., 2004), eigenshape analysis (Unal et al., 2008) and Markov random fields (Paulsen and Hilger, 2003). More broadly, shape analysis of the external ear, excluding the canal itself, has been investigated for biometric security purposes using Poisson disk sampling (Sun et al., 2014), geometric edge detection (Omara et al., 2016), local surface variation descriptors (Zhang et al., 2017), local binary pattern descriptors (Abaza and Harrison, 2013) and convolutional neural networks (Cintas et al., 2017).

Convolutional Neural Networks (CNN) (Krizhevsky et al., 2017) gained popularity within the medical imaging community for classification and segmentation tasks (Tajbakhsh et al., 2016). Research relevant to the human ear includes anatomical landmark detection in 2D images (Cintas et al., 2017), using ConvNets (Lecun et al., 1998); classification of otitis media in otoscopy 2D images (Sundgaard et al., 2021), using the Inception CNN (Szegedy et al., 2016); classification and landmark localization of CT images for cochlear implants (Zhang et al., 2020) and vestibule segmentation from CT images (Zhang et al., 2021), using the 3D voxel U-Nets (Çiçek et al., 2016); segmentation of the intra cochlear anatomy in CT images (Wang et al., 2019a) and detection of metal artifacts in cochlear implant CT images (Wang et al., 2019b), using 3D voxel GANs (Goodfellow et al., 2014).

Although the aforementioned studies have explored how to identify landmarks or classify objects in the ear for diagnostic purposes, methods to improve the efficiency and accuracy of customized hearing aid fits are lacking. Herein we describe how CNNs can be used to perform registration, classification, and segmentation of ECIs. These tasks are laborious and usually performed manually by hearing aid design experts. Tools to increase the efficiency of hearing aid design are urgently needed because of the expected increased use of customized hearing aids (WHO, 2018). The PointNet architecture (Qi et al., 2017a) is compatible with the 3D point images that result from 3D scanning ECIs. PointNet CNNs, see Fig. 1, enabled deep learning using 3D point clouds for which 2D image-derived methods cannot be directly used due to the inherent data irregularities; and due to their simplicity and strong representation ability, they have been used as building blocks for several applications (Guo et al., 2021). Key benefits include: (a)

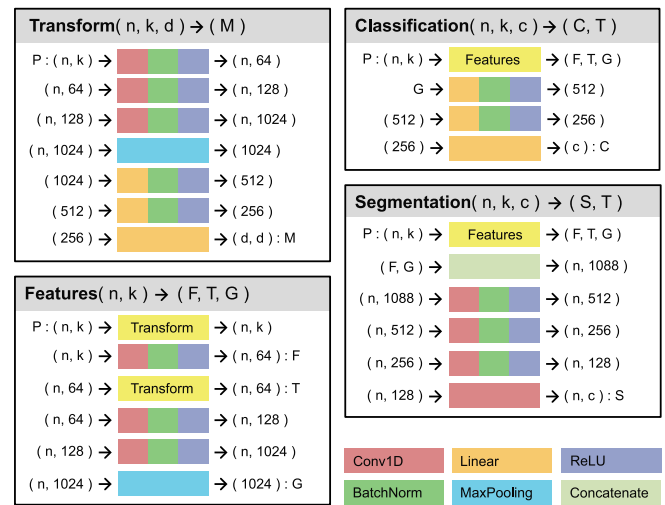


Fig. 1. The PointNet architecture represented as functional block diagram, where n is the number of points, k is the points' dimensions, d is the transformation matrix dimensions, c is the number of classes. The Transform block inputs the point cloud P and returns the transformation matrix M . The Features block inputs the point cloud P and returns the features' tensor F , the transformation matrix T and the global feature G . The Classification block inputs the point cloud P and returns the class C value and the transformation matrix T . The Segmentation block inputs the point cloud P and returns the per point class list S and the transformation matrix T .

they are computationally efficient compared to 3D voxel CNNs, (b) they do not require point data to be consistently oriented because they co-train Spatial Transform Networks (Jaderberg et al., 2015) to solve the transformation invariance problem; (c) they handle the permutations' problem, whereby the order of the input points supplied does not matter (Qi et al., 2017a). While PointNets have been used for medical image analysis, such as anatomical shape analysis (Gutierrez-Becker et al., 2021), tractogram filtering (Astolfi et al., 2023), white matter analysis (Xue et al., 2023) and orthognathic surgical planning (Ma et al., 2023), there are no precedents relevant to ECI analysis available. The aim is therefore to establish a baseline study on the use of 3D point CNNs for audiology which motivates the use of the foundational PointNets as opposed to more complex architectures such as hierarchical PointNet++ (Qi et al., 2017b), Point Attention Transformers (Yang et al., 2019) and Structural Relational Networks (Duan et al., 2019). The article presents three topics of analysis in the following sections, namely ECI (a) handedness classification, (b) registration in consistent orientation, and (c) semantic segmentation by removal of irrelevant points. Until now, all three topics have been performed manually by hearing aid designer experts.

3. Materials and methods

The ECIs received, upon obtaining research board approval (CIRB 2020/3140), were sourced from four different laboratories. The ECIs were created using different 3D scanning equipment and post-processing software resulting in a variety of file formats including (a) stereolithography (STL) meshes, (b) proprietary Magics3D (MGX) meshes, and (c) comma-separated value (CSV) 3D point clouds. Only 3000 from a total of 10,000 ECIs were processed by the authors and used in this study due to the labor intensity involved, requiring circa three months to perform data corrections and label the ECIs manually. Files were converted to STLs due to the format's simplicity. Point clouds were converted to meshes using MeshLab's (Cignoni et al., 2008) surface reconstruction algorithm. File format normalization resulted in a dataset ($n=3000$) of unprocessed ECIs, denoted as **U**. Those were further processed, including both automatic and manual operations, resulting in a separate dataset ($n=3000$) denoted as **P**.

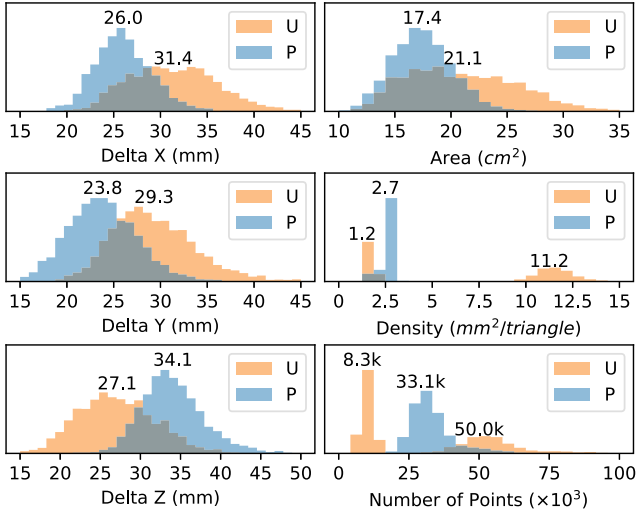


Fig. 2. Geometric characteristics of unprocessed [U] and processed [P] ECIs. The variation of processed ECIs is lower than unprocessed.

Automated tasks included removing disconnected surface domains, closing mesh holes, removing non-manifold edges, orienting vertex normals coherently, decimating faces using quadric edge collapse (Garland and Heckbert, 1997), or refining faces using Loop subdivision (Boyé et al., 2010) aiming to achieve consistent point density per surface area, see Fig. 2. Manual tasks included: annotating ECIs as being left or right; orienting ECIs from the arbitrary 3D scanner's coordinate system into a consistent frame of reference, where the Z-axis is aligned vertically along the direction of a standing person, and the Y-axis pointing towards the depth of the ear canal; and clipping ECI models to remove irrelevant parts of the silicone mold and reveal the ear canal's cavity. Unlike the automated geometric and topological operations, the manual tasks were semantic and required domain expertise. Here we report the development of methods aiming to accurately automate these semantic operations.

The dataset was serially numbered and the first 2048 ECI were used for training while the remaining 952 (30.7%) were held out for testing. The dataset was not randomized before splitting because a preliminary inspection of the data revealed several cases where consecutive ECIs appeared visually similar. Therefore, randomization was performed only after splitting the dataset to prevent ECIs from presumably the same patients appearing in both the training and test sets. Before point sampling, the ECIs were translated from their geometric center to the origin and normalized using a uniform scaling factor (0.01). The CNNs were implemented using the PyTorch framework and trained using the Adam optimizer with an initial 0.01 learning rate. An Intel i9-12900K CPU with an RTX3080 12 GB GPU workstation was used for training.

4. Classification

Classifying ECIs as left or right is necessary before performing comparisons. From the total 3000 ECIs examined, 1516 (50.5%) and 1484 (49.5%) belonged to left and right ears, respectively. Most of the ECIs received, 2599 (86.6%) were already labeled with the letters “L” or “R” in their file names. Nevertheless, they were all manually cross-checked to establish the ground truth. Only 5 (0.2%) were found to be mislabeled, using the wrong letter, by the person who created the 3D scan. Despite being laborious, labeling ECIs is not difficult, but it does require familiarity with the outer ear's anatomy. Orienting the geometry visually on-screen such that handedness becomes easier to discern is helpful, see Fig. 3 for a pair of unprocessed and processed ECIs.

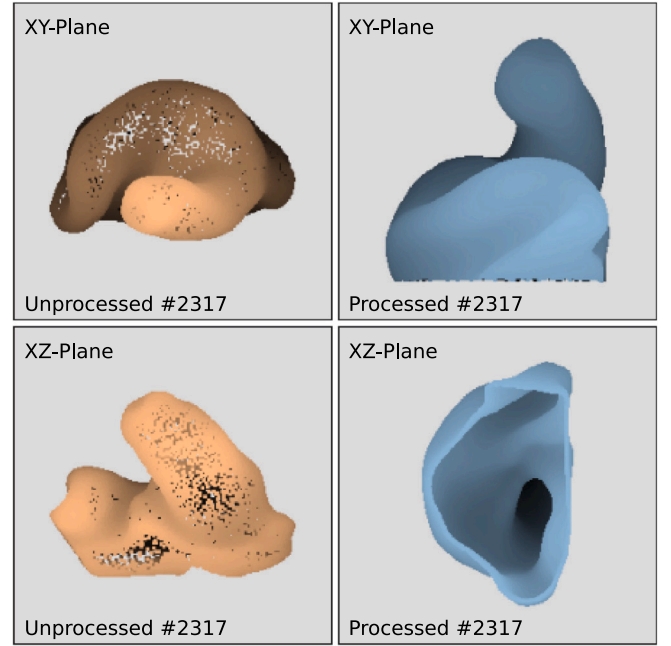


Fig. 3. Unprocessed ECIs are often closed solids that were manually clipped and oriented such that XZ-plane is the front view of the ear. It is easier to conclude that the depicted ECI belongs to a right ear by inspecting its processed version.

Geometric analysis was initially employed to determine the ECIs' handedness using semantic features of the ear. When viewed from outside, normal to the XZ-plane, the ear canal's direction as it progresses along the depth Y-axis is consistent and provides a very good indication of the ear's handedness. Moreover, the C-shaped region near the entrance of the canal, known as the concha, provides another strong hint of directionality. Therefore, two criteria named “depth” and “normal” were defined and computed as follows: Provided with a processed ECI, its mesh was contoured towards the Y-axis every 0.5 mm, retaining only single closed loops. The depth criterion uses the vector from the outermost contour's center, to the center of the contour three quarters in the ear canal. The sign of this vector's dot product with the X-axis, renders an ECI as left if negative and right otherwise. The depth ratio was empirically selected to avoid contours towards the end of the ear canal because they are generally distorted by the foam block used during silicone injection. Contours up to one-quarter into the canal were examined for computing the normal criterion. The contours were first decimated by removing consecutive points closer than 5 mm and cubic B-Spline curves were fitted to simplify and smoothen their shape. Normal vectors were evaluated at 256 locations along the curve's parametric domain, spaced evenly in the arc-length sense. The mean normal direction was thereafter used as an indicator of the ear's handedness; again using the sign of its dot product with the X-axis. Geometrically this generally works because for an asymmetric C-shaped curve, the number of normals pointing towards the positive versus the negative X-direction is unequal, while the largest number of normals points towards the direction of bend, see Fig. 4.

Overall, the criteria evaluated over all 3000 ECIs yielded 99.3% and 97.6% accuracy for the depth and normal criteria, with only 22 and 72 incorrect classifications, respectively. Combined, 97% passed both criteria, 2.9% exclusively failed either, and 0.1% failed both. The cause of failure, see Fig. 4 for examples, is mainly due to ECI alignment and clipping issues. Thus, while geometric classification performed well, its success depends on the properties of fully processed ECIs, namely proper registration and segmentation. In addition, several empirically

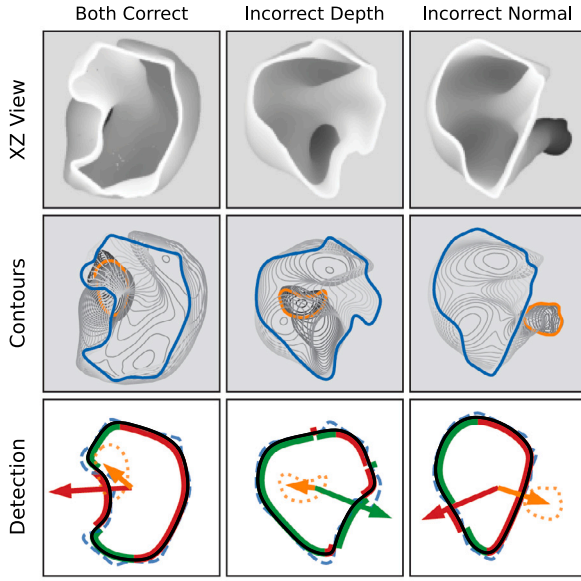


Fig. 4. Geometric classification samples. The orange inner contour is used for the depth criterion. The blue outer contour is approximated with a B-Spline curve and its normals and mean normal vectors are color-coded as green for right and red for left ear directions.

tuned parameters, such as contour spacing and selection, render this approach rather fragile.

The shortcomings of the geometric algorithm motivated the development of a CNN capable of classifying unprocessed ECIs. The input to the CNN is a tensor of 2048 points and its output is the Negative Log Likelihood (NLL) of the point cloud belonging to a left or right ear class. ECIs contain tens of thousands of points each, which allows for augmenting the training dataset by randomly sampling the same geometry multiple times. Additional data augmentations were performed using random rotations and translations within $\pm 180^\circ$ and ± 5 mm per coordinate axis. Each ECI was sampled 16 times, after shuffling its points, and transformed randomly 16 times, totaling 256 replications. Neural networks are typically overconfident (Kristiadi et al., 2020), which prevents their Soft-Max output probabilities from being a dependable indicator of prediction confidence (Guo et al., 2017). Since ECIs contain more than 2048 points each, instead of classifying each ECI once, we performed 256 evaluations, with 16 randomized groups of 2048 points, times 16 random transformations, to obtain a prediction consensus score.

Several CNNs were trained, starting with using only two and doubling the number up to 2048 ECIs, to understand the relationship between the number of ECIs used for training versus the model's accuracy. The same remaining 952, unseen during training, ECIs were used for testing all the models. In addition, the CNNs' performance in classifying processed versus unprocessed data, was evaluated using combinations of the **U**, **P** datasets, where **PP** are models trained and tested with processed data, **UU** were trained and tested using unprocessed data, **UP** were trained with unprocessed data and tested using processed data and **PU** were trained with processed data and tested using unprocessed data. The number following each model represents the number of ECIs used for training. Models trained with a limited number of ECIs performed as expected with nearly 50% accuracy, while all models with the maximum number of ECIs allowed, reached above 95% see Fig. 5a. The results of combining the **U** and **P** datasets are also to be expected: using processed data is beneficial but not strictly required. The sharp increase from nearly random to over 95% for all models when trained with at least 256 ECIs is, however, surprising. The **UU** and **PP** models achieve over 95% with 128 and 64 ECIs, respectively. The consensus score expresses how certain the CNN is about

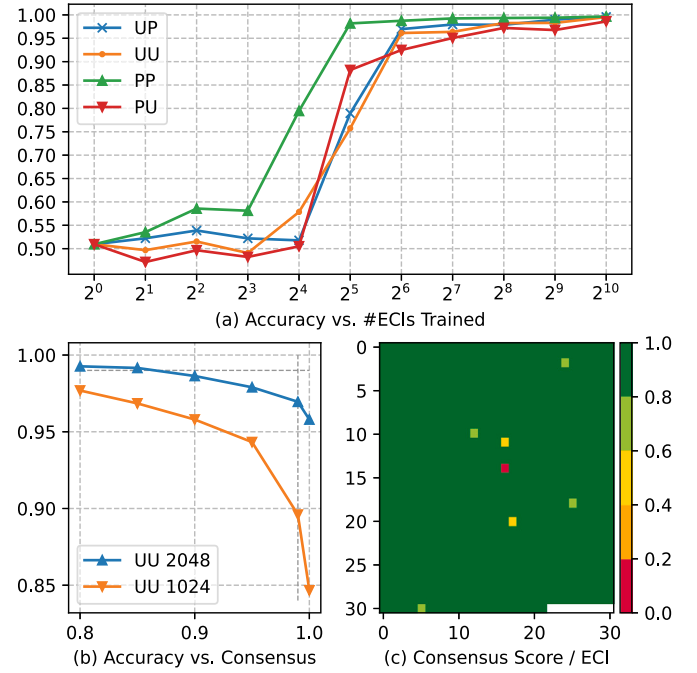


Fig. 5. (a) Test accuracy versus the number of ECIs used for training. (b) Accuracy versus consensus for models trained and tested with unprocessed data with 2048 and 1024 input points, with dashed lines at 99%. (c) Matrix of consensus scores of all ECIs tested using UU2048.

the handedness of an ECI, in the sense of how many times it predicted the same ECI being left or right, over the total of 256 predictions, evaluated under different point groups and transformations. Adjusted model accuracy was defined as the mean of accuracies per ECI greater or equal to the consensus threshold. For example, if 100% consensus per an ECI is required, that is all 256 predictions to be equal, then the accuracy of the UU2048 model is 95.8%. However, with 99%, 95%, and 90% consensus thresholds, the associated adjusted accuracies become 97%, 97.9%, and 98.6%, respectively, see Fig. 5b. Finally, the consensus scores of all 952 tested ECIs, see Fig. 5c, highlights that the UU2048 model misclassified only one ECI with 18.8% consensus score and an additional one with 40.6%, while all others fell in the range of circa 60% and above. Visually inspecting the worst results presents some interesting aspects of the model's behavior, see Fig. 6. ECI #2506 is unusually elongated in the vertical direction with a very narrow canal cavity while #2506 and #2693 contain excessive amounts of silicone. Contrastingly, #2991 is either incomplete or trimmed too short. These results are important because they highlight that low consensus scores are associated with highly unusual ECIs. We foresee, therefore, that this approach may be also used as a quality control check on ear canal scans.

Overall, the results using both the geometric and the CNN methods were positive. In contrast to the geometric method, which relied on properly aligned and clipped impressions, explicit semantic features and hand-tuned parameters, CNNs were robust and arrived at respectable results when trained on a modestly sized dataset of labeled samples. This outcome motivated the development of a CNN capable of automatically orienting ECIs.

5. Registration

Geometric registration recovers the transformation required to bring a shape as close as possible to another. The objective is to minimize the effects of rotation and translation. Registration aids feature detection, and it is used when computing pairwise similarities (Audette et al., 2000). The transformation may be rigid (Darkner et al., 2007), where

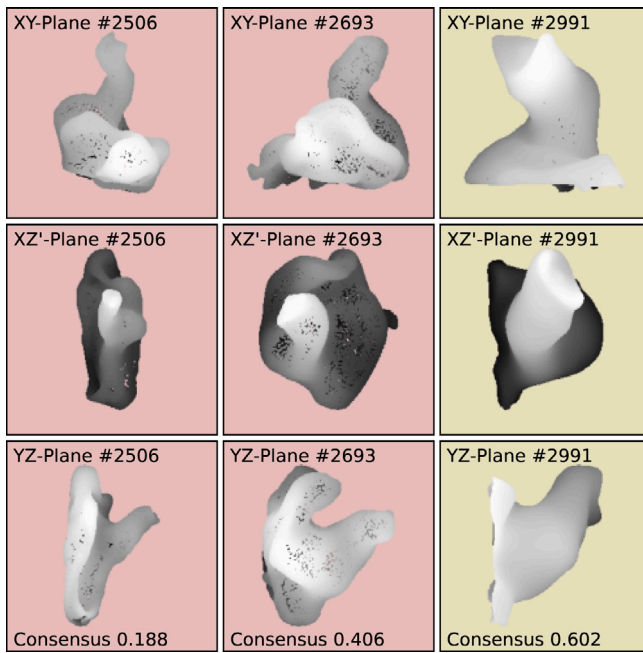


Fig. 6. Worst classification results, where the first two columns present misclassified instances and the third an ECI correctly classified but with low consensus score.

only rotation and translation are allowed, or non-rigid (Darkner et al., 2008; Andresen and Nielsen, 2001). Iterative Closest Point (ICP) is a staple pairwise rigid registration method (Besl and McKay, 1992) although discrepancy minimization methods, based on Optimal Transport theory (Mérigot et al., 2018; Alvarez-Melis and Fusi, 2020) as well as neural networks (Wang and Solomon, 2019; Choy et al., 2020), have been recently studied. Registration methods are highly sensitive to the initial configuration of the shapes provided; if the shapes are insufficiently close, the results are highly unsatisfactory (Zhang et al., 2022). The goal of the ECI registration process presented here is to orient them for consistent visualization and to preset the shapes such that the initial conditions are favorable for subsequent analysis using the aforementioned fine-grained pairwise registration methods.

We established the convention where the front view of an ear is in the world XZ-plane. All 3000 ECIs were aligned using this convention manually. The transformation matrices from P to U were recovered using the ICP method. Evaluating several ICP variants produced unsatisfactory results, even though the same points from the same meshes were used. The methods examined included the original point-to-point (O-ICP) by Besl and McKay (1992), the Anderson acceleration point-to-point (AA-ICP) by Walker and Ni (2011), the fast point-to-point (F-ICP) by Zhang et al. (2022), the robust point-to-point (R-ICP) by Zhang et al. (2022), and the point-to-plane (P-ICP) by Chen and Medioni (1992). To overcome this obstacle we developed the Q-ICP variant, which uses the 24 orthogonal rotations of a cube, namely six faces times four 90° rotations, to rotate ECIs before invoking the F-ICP method. The transformation producing the lowest pairwise point-distance Root Mean Squared Error (RMSE) was selected. An ECI was considered correctly recovered when $RMSE \leq 1$ mm. While theoretically, the method should be 24 times slower than F-ICP, in practice it was only circa 12 times and 9 times from the original ICP method. This is due to early termination when the RMSE is below the acceptance threshold and because unprocessed ECIs were not completely randomly oriented in the first place. Despite being the slowest, Q-ICP recovered 100% of the transformation matrices, see Table 1.

An ECI registration CNN was trained to predict the rotation required to orient an unprocessed ECI to its desired state, using the recovered transformation matrix information collected from the manual

Table 1

Transformation recovery with various ICP methods.

Method	Recovered	Rec. rate	Time (s)	Rel. time	# Iters/ECI
O-ICP	234	7.8%	0.059	1.00	108
AA-ICP	243	8.1%	0.488	8.23	141
F-ICP	295	9.8%	0.045	0.76	57
R-ICP	363	12.1%	0.253	4.27	285
P-ICP	374	12.5%	0.511	8.60	173
Q-ICP	3000	100%	0.539	9.08	561
H-ICP	950	99.8%	0.017	0.28	9

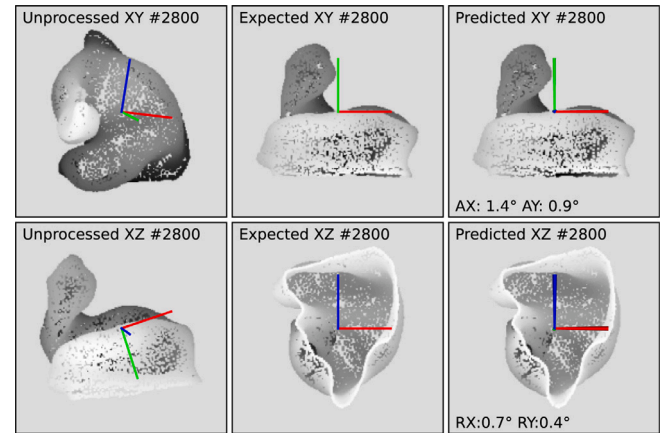


Fig. 7. Original, expected and predicted ECI transformation. The colored lines represent the desired basis where its X, Y and Z vectors are red, green and blue, respectively. The darker lines represent the 256 predicted vectors. AX, AY are the angle errors between predicted and expected vectors and RX, RY are the standard deviations from the mean.

alignment process performed. Essentially, the CNN emulates 3D view manipulation actions of a computer-aided design operator, who has a visual understanding of the ear's anatomy. This form of registration may be considered as a “singular” or “unitary”, as opposed to pairwise, because it does not rely on a secondary shape. The CNN was trained using 2048 unprocessed ECIs with the remaining 952 used for testing. Its input was a tensor of 2048 ECI points and its output was the X and Y basis vectors of the 3D rotation matrix. The CNN's architecture and training regime was the same as for classification, but with the last layer Soft-Log-Max activation omitted, and the Mean Squared Error (MSE) loss function used instead. Therefore, these modifications investigate whether PointNets may be employed for regression tasks. An example of an unprocessed ECI, its processed orientation and predicted transformation is illustrated, see Fig. 7, where the basis vectors are always in reference to the desired orientation.

Training the CNN with the MSE loss against the 6-tuple, from the X and Y axes of the recovered rotation matrices, does not constrain the solution to a proper rotation, where its basis vectors must be orthonormal. Thus, to recover the rotation we computed the Z-axis, as the cross-product of the X and Y axes, and enforced orthogonality, using the cross-product between the Y and Z axes, followed by normalization. To derive the predicted transformation the CNN was randomly sampled with replacement 256 times using different groups of points from the same ECI in conjunction with applying random transformations per group. The mean basis vectors were used to derive the rotation and the standard deviation from the mean as a measure of prediction quality.

The angular errors between the expected and the predicted basis vectors were evaluated, see Fig. 8. Effectively, only 2 ECIs were oriented in the completely wrong direction with angular errors above 90°. From the 952 ECIs tested, 99.4% were below 45° from target in both directions. The mean Y-axis error was 6.50° (6.29) with 99.2% of ECIs below 19° error, whereas the mean X-axis error was 10.68° (8.89) with 99.2% of ECIs below 32° error. The errors between the X and

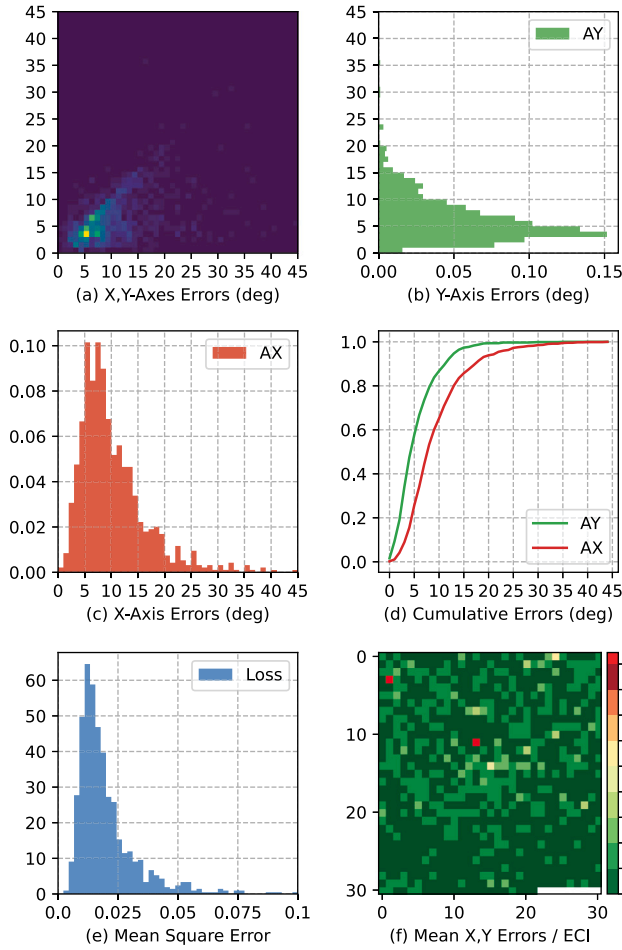


Fig. 8. (a) Map of axial error distribution for both X and Y axes, (b) Distribution of errors for the X-axis, (c) Distribution of errors for the Y-axis, (d) Cumulative axial error distributions for both X and Y axes, (e) Distribution of MSE test loss, (f) Map of mean axial error for all ECIs tested.

Y axes present an asymmetry, with the Y-axis being more successfully predicted than the X-axis. This behavior may be attributed to prioritizing the Y-axis when orthonormalizing the results, but perhaps this was already a trend in the data because from experience it is often visually easier to discern the ear's depth direction versus its upright direction. Examining the standard deviation of the predicted X and Y vector angles against their mean directions was useful for detecting the outliers. Angle variation in the X and Y axes against the predicted means was overall 1.07° (1.47) and 0.73° (0.67), respectively. Using one standard deviation above the mean as a threshold, corresponding to circa 2.5° and 1.5° , was used to identify problematic results. These thresholds yielded a 95.3% accuracy in assessing whether the predicted rotation was potentially wrong, with 12 true positives, 20 false positives, 25 false negatives, and 895 true negatives.

The CNN often appeared to produce better ECI orientations when comparing expected versus predicted pairs. All predicted transformations were therefore visually inspected to understand the CNN's behavior and to qualitatively assess the results. Indeed, several ECIs showed improvement in orientation, see Fig. 9 for indicative examples. Overall, 96.1% of transformations appeared adequate with 12.5% better, 83.6% same, 3.7% worse than expected and 0.2% completely wrong. It is likely the neural network identified the broader trend of how ECIs were generally oriented manually and for ECIs that were outside the broader mean, it was able to propose a rotation closer to the trend's mean. Finally, to assess the CNN's performance as a pre-alignment method for pairwise registration, the transformation matrix recovery process

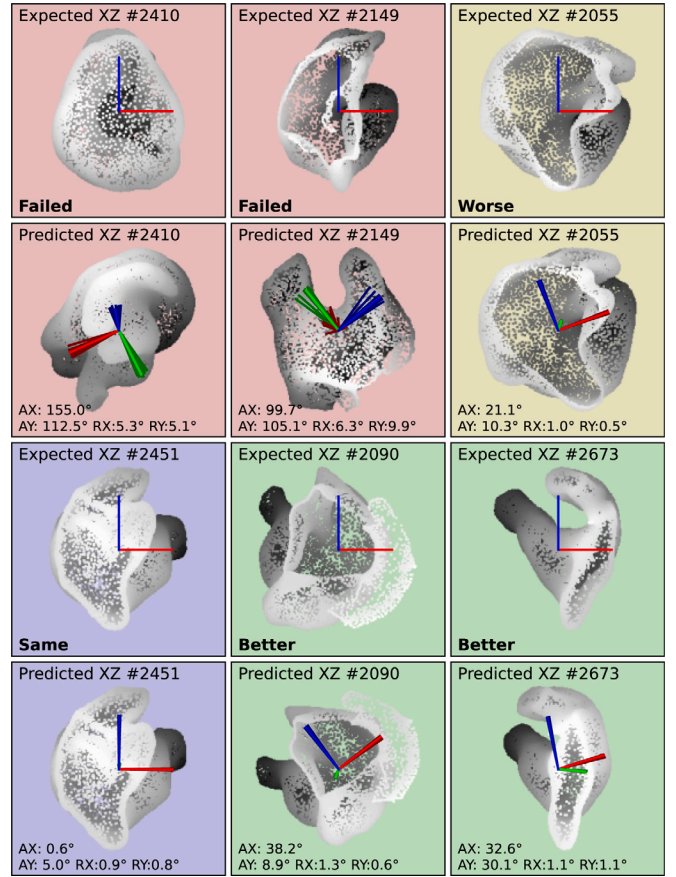


Fig. 9. Qualitative analysis of the predicted versus expected orientations categorized as failed, worse, same, and better. The only two failed cases are presented, an example where the orientation was unsatisfactory, an example of a prediction very close to expectation, and two instances where the CNN improved the orientation. Except for the two failed cases, all predicted orientations were plausible. The CNN surprisingly corrected several mistakes made during manual orientation.

was performed using the CNN's prediction before executing the F-ICP method. The hybrid or H-ICP entry, see Table 1, recovered 99.8% of the ECIs and improved F-ICP's performance, which required an average of only nine iterations before converging. The time reported, 0.017 s does not include the CNN's 0.131 s, for all 256 evaluations per ECI.

The registration CNN was able to accurately orient ECIs without a template. In some cases, the results were better than a human-specialist operator. We were surprised that the neural network could correct our own mistakes. Although a template was not used in the pairwise sense, the CNN may be considered as the mean shape orientation, and therefore the template itself. The reasons for failed orientations and worse results are unclear; we note that all those ECIs contained unusual features, such as excessive amounts of silicone, closed volumes, and 3D scanning artifacts. However, the network was able to successfully orient many others with similar characteristics.

6. Segmentation

The objective of ECI segmentation was to remove semantically irrelevant parts from the outward-facing surface of the ear mold. Primarily this was aimed at automatically clipping watertight ECI meshes to reveal the ear canal's cavity. Nevertheless, the exterior regions of most ECIs were generally inconsistently scanned and/or meshed. This is because molds are scanned with the ear canal's direction facing towards the sensor; the opposite side thus suffers from lower data quality. Therefore, the secondary aim was to eliminate exactly those problematic

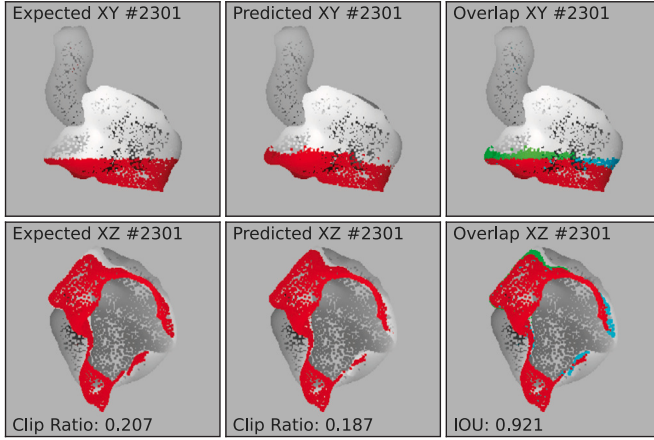


Fig. 10. Expected versus predicted segmentation result. The overlap column presents correctly retained and removed points color-coded as gray and red, respectively. Incorrectly retained and removed points are color-coded as cyan and green, respectively.

areas. The task was approached as a per-point classification process, where points are labeled as retained, for being part of the ear canal or removed, for being artifacts of the silicone ear mold. Labeling was performed by comparing the unprocessed and processed ECI datasets. An unprocessed point was retained if there existed a non-empty ball of radius 0.5 mm containing points from its processed counterpart. This was necessary because the processed ECIs were adjusted for point density. Effectively, a CNN was trained to replicate the qualitative judgment of a human expert. Its architecture was based on the semantic segmentation variant of the PointNet. Its input was a tensor of 2048 points and its output was the likelihood per point of being retained or removed. To evaluate the results the Intersection Over Union (IOU) criterion was used, defined as $IOU(E, P) = \frac{\sum(E \cap P)}{\sum(E \cup P)}$, where E and P are the expected and predicted classes per point, see Fig. 10 for an example.

All ECIs were clipped plausibly with no obvious mistakes. The CNN removed points exclusively from the outermost faces of the mold. Overall, the CNN's mean IOU was 0.934 with 95% of the ECIs over IOU 0.8, see Fig. 11a, b. In addition, the clipping ratio, defined as the number of points removed from its total number of points, between the expected and predicted results was examined. Interestingly, segmentation results presented a strong bias towards under versus over clipping, with 20.8% and 11.0%, respectively; considering those within 5% difference as within tolerance, see Fig. 11c.

Visual inspection of all predicted segmentations confirmed the bias towards under-clipping, see Fig. 12 ECI #2859. This was problematic for some ECIs, where parts of the outer silicone surface were unnecessarily retained. This behavior may be attributed to the 0.5 mm labeling tolerance used and the presence of high-quality unprocessed ECIs, 10% requiring no clipping at all, such as #2770. Completely unclipped ECIs were also the reason the IOU distribution exhibits a sharp edge in its top bin. The lowest IOU 0.432, represents a very conservative segmentation for perhaps the most aggressively clipped ECI #2639. Nevertheless, while the CNN was conservative, it still produced a reasonable clipping. In several instances, the predicted segmentation appeared better than the expected manual result. Improved clipping, occurred when the proposed solution adjusted the notional clipping surface's offset to include additional irrelevant points compared to those selected by hand, such as #2323, or when its orientation was adjusted to better align it with the exterior face of the mold, such as #2485. After visually evaluating all ECIs tested, 91.5% of the ECIs were adequately segmented, with 10.5% showing improvement and only 8.5% being worse.

A consensus-based approach was used for clipping points from ECIs. The segmentation CNN was evaluated several times using random

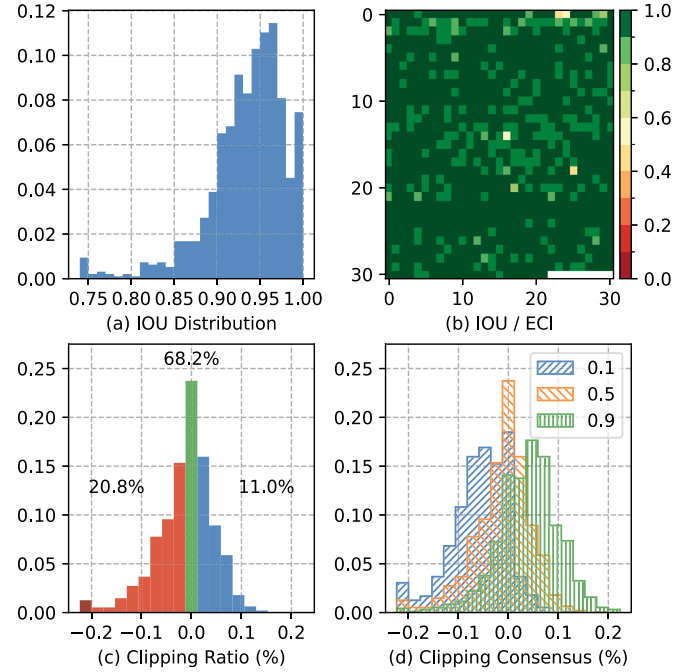


Fig. 11. (a) Distribution of IOUs, (b) Map of test dataset IOUs, (c) Distribution of clipping ratios, and (d) Clipping ratio distributions for three consensus score thresholds.

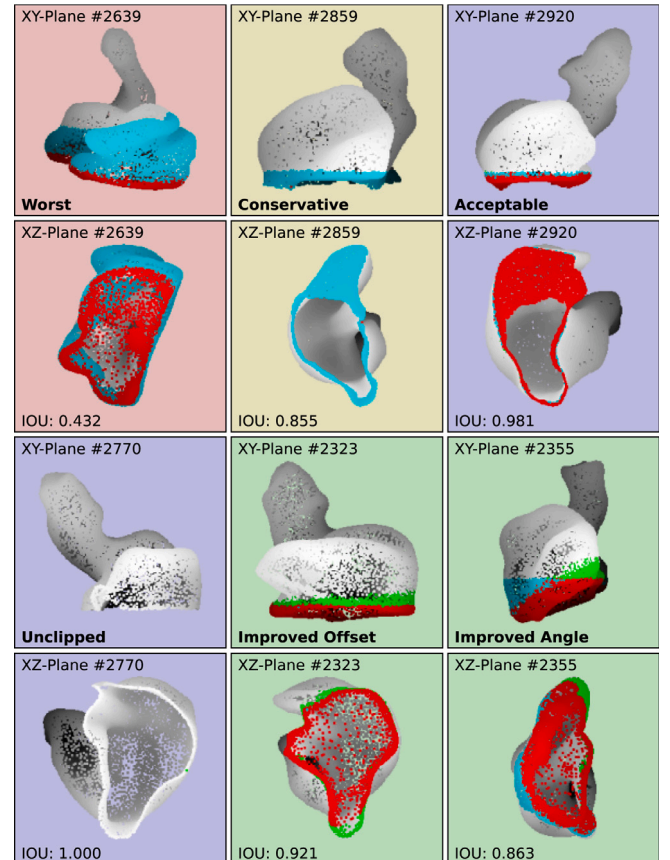


Fig. 12. Examples of expected versus predicted ECI results qualitatively categorized. The worst performing ECI is still reasonably clipped. Conservative clipping retains unnecessary surface fragments. High-quality ECI were completely unclipped. Improved clipping occurred when removal was expanded to eliminate more irrelevant points and when the clipping direction aligned better with the ECI.

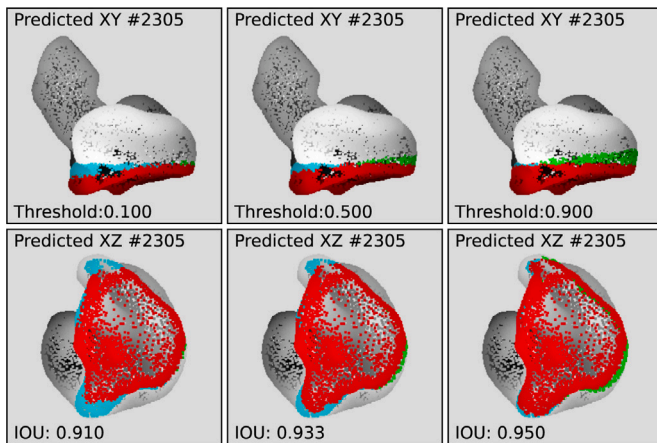


Fig. 13. Low consensus threshold 0.1 results in conservative point removal while high consensus threshold 0.9 yields aggressive point decimation. Cyan points were removed by us but retained by the CNN, while green was retained by us but removed by the CNN.

groups of 2048 points per ECI until at least ten votes per point were available. The consensus score conceptually captures the CNN's confidence in retaining or removing points. Using a threshold value, allowed for compensating the under-clipping bias and adjusting the clipping ratio distribution, see Fig. 11d. Setting the threshold to 0.5 produced the previously explained default behavior, however setting lower and higher values promoted under and over clipping, respectively. This enabled controlling how conservative or aggressive the CNN behaved. An example of segmentation results for three threshold levels is presented in, see Fig. 13. It also illustrates that the threshold value controls the shape of the CNN's notional clipping surface. Low thresholds removed points from a tight region near the exterior of the ECI while higher values completely changed the clipping shape's orientation and offset. Setting the threshold to 0.9 and visually reassessing the segmentation results reduced the number of unsatisfactory ECI clippings from the default case of 8.5% to 3.1% and increased those improved from 10.5% to 22.3%.

Semantic segmentation using CNNs was effective in removing artifacts from the 3D scanned ECIs. Remarkably, the CNN learned to clip ECIs' exactly about the problematic areas and with a plane-like surface, as it was performed manually. It also often produced better segmentations than manually performed. The use of the consensus score threshold allowed for some degree of tuning its behavior. It is also foreseeable that the same per-point classification method may be extended to categorize points in an anatomical sense should we label the ECIs using more classes.

7. Discussion

The methods presented offer solutions for automating the pre-processing of ECIs to assist in the analysis and design of hearing aids. This considers that the operations performed manually require expert knowledge and several weeks of work, whereas using the methods presented requires a few minutes. Additionally, the tasks were transformed from a process of geometric manipulation to one of visual inspection. PointNet CNNs were evaluated for both classification and segmentation tasks, for which they were designed, as well as for registration. While achieving above 90% accuracy for all tasks, in a population of 10,000 ECI, inspecting those with the lowest accuracies still requires substantial effort. This motivates future investigations where additional geometric features per point may be incorporated, such as the normal direction and local curvature characteristics, or the use of more advanced network architectures capable of capturing local structures, to improve the results. Sampling the CNNs multiple times

provided the means for partially overcoming their overconfidence, and highlighted unusual ECIs as well as human errors in the original dataset. Nevertheless, these were conflated with CNNs mispredictions. Interpretability is a fundamental limitation of deep learning black-box models (Fan et al., 2021). Finally, the dataset was regionally sourced and thus represents demography from the South East Asia. Therefore, the generality of the methods and results presented against a broader demography cannot be assessed.

8. Conclusion

We presented a study deploying CNNs for classification, registration and segmentation of ECIs with positive results. Those are plausible in both cases where an objective criterion is available, such as in the case of handedness classification, but also even when the model is called to emulate human vision tasks, such as the registration and segmentation scenarios. What is remarkable about the models is that they were capable of performing these tasks using unprocessed ECIs featuring a broad range of quality characteristics, including ECIs with uneven point densities, arbitrary orientation, excessive amounts of silicone, closed, open or fragmented boundaries, and various degrees of silicone penetration into the ear canal. These are key issues that complicate geometric analysis. In conclusion, there are numerous opportunities for further analysis of the ear canal's morphology, as well as quality control, using the tools developed and/or extending their capabilities to discern anatomical features, deducing the number of bends and geometric properties of the canal, discarding or even correcting ECIs with incomplete features. We hope the development of such tools will pave the ground for potentially improving the design, performance and comfort characteristics of hearing aids in the near future.

CRedit authorship contribution statement

Stylianios Dritsas: Data curation, Methodology, Writing – original draft. **Kenneth Wei De Chua:** Data curation, Supervision. **Zhi Hwee Goh:** Data curation, Investigation. **Robert E. Simpson:** Investigation, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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